

VPS Production Network Segmentation

Architecture Design & Operational SOP (Public / Redacted Edition)

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Category: Cybersecurity / IT Infrastructure & SecOps

Note: This is a redacted, portfolio-safe edition. All real IP ranges, VLAN IDs, ISP names, device port mappings, and internal interface names have been replaced with placeholders. The unredacted operational SOP is maintained privately and is not published.

1. Purpose

This document describes the standard process used to design, implement, operate, monitor, and maintain a segmented production network for a virtualized (Proxmox-based) VPS environment. It demonstrates an approach to isolating management, production virtual machine, and storage traffic while maintaining high availability, redundancy, and operational consistency.

2. Scope

This document is written for infrastructure engineers, network engineers, security engineers, SOC analysts, and system administrators. It covers, at a conceptual level:

- A next-generation firewall
- A managed core switch
- A secondary/out-of-band management switch
- A multi-node hypervisor cluster (Proxmox VE)
- Production virtual machines
- A dedicated storage replication network

3. Roles & Responsibilities

Role	Responsibility
Infrastructure Engineer	Deploy and maintain hypervisor infrastructure
Network Engineer	Configure VLANs, firewall, and switching
Security Team	Review firewall rules and security controls
SOC Team	Monitor logs, alerts, and security events
Change Manager	Approve production changes
IT Manager	Final approval of infrastructure modifications

4. Architecture Overview

At a high level, the environment is structured as follows:

Internet
|
Redundant ISP Uplinks (Primary / Secondary)

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|
Edge Firewall
|
Core Switch
|
Hypervisor Cluster (3 nodes)
|
Segmented VLANs (Management / VM / Storage)

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Design goals:

- Dual-ISP redundancy for internet connectivity
- VLAN-based network segmentation
- LACP-bonded uplinks for resilience and throughput
- Secure, isolated management access
- Dedicated, WAN-isolated storage replication network
- Controlled, NAT-based internet access for production VMs

5. Network Segmentation

Segment	Purpose
Management	Firewall, hypervisor, and out-of-band (IPMI) management access
VM Network	Internet-facing production virtual machines
Storage	Cluster / storage replication traffic only

Security Principles

- The Management segment must never be directly Internet-accessible.
- The Storage segment must never route to the WAN.
- The VM segment reaches the Internet only through controlled NAT / firewall policy.

6. Hardware Inventory (Categories)

Device	Role
Edge Firewall	Perimeter security / routing
Core Switch	VLAN trunking and distribution
Management Switch	Out-of-band SOC / admin access
Hypervisor Node 1–3	Virtualization cluster

Specific vendor models, serial numbers, firmware versions, and asset tags are omitted from this public edition.

7. IP Addressing Approach

Each network segment is assigned its own private, non-overlapping RFC1918 subnet with a dedicated gateway. Actual subnet ranges, gateway addresses, and host IPs are considered sensitive operational data and are withheld from this public edition.

Segment	Addressing Scheme (example / placeholder)
Management	10.x.x.0/24 (placeholder)
VM Network	10.x.x.0/24 (placeholder)
Storage	10.x.x.0/24 (placeholder, non-routable to WAN)

8. Port & Interface Mapping

The private SOP documents exact physical port assignments on the firewall and switches, plus real interface names on each hypervisor node. These mappings are internal operational detail and are withheld here. Conceptually, the design reserves:

- Dedicated WAN ports for each ISP uplink
- A trunked port carrying the Management VLAN
- A high-speed trunked uplink carrying VM and Storage VLANs
- Reserved ports for out-of-band (IPMI) management
- Reserved ports for SOC / administrative access

9. Implementation Procedure (Overview)

Step 1 — Firewall Configuration

- Configure WAN interfaces and failover/SD-WAN policy.
- Create Management, VM, and Storage VLANs.
- Configure NAT for the VM segment.
- Explicitly block Storage-segment traffic from reaching the WAN.
- Restrict firewall administration to the Management segment only.

Step 2 — Switch Configuration

- Configure out-of-band management ports.
- Configure access ports for management devices.
- Configure trunk ports carrying the required VLANs.
- Configure LACP link aggregation on uplinks.
- Verify VLAN tagging end-to-end.

Step 3 — Hypervisor Configuration

On each cluster node, create a bonded interface, a VLAN-aware bridge, and per-segment VLAN sub-interfaces. Assign management, storage, and VM-facing addressing per the (private) IP plan. Verify GUI access, cluster health, and storage connectivity before proceeding.

10. Operational Traffic Flows

10.1 VM Internet Traffic

VM → VM VLAN → Core Switch → Firewall → NAT → Internet

10.2 Storage Replication

Node 1 ↔ Node 2 ↔ Node 3 (Storage VLAN only, isolated from WAN)

10.3 Management Traffic

Administrator → Management VLAN → Firewall → Switch → Hypervisor GUI

11. Verification Checklist

Check	Expected Result
Node-to-node connectivity	Success
Hypervisor GUI	Accessible from Management segment
Out-of-band management	Accessible from Management segment
VM Internet reachability	Success via NAT
Storage sync	Healthy
Storage segment WAN isolation	Blocked (confirmed)
SOC access	Successful, scoped to Management segment

12. Monitoring

Continuously monitored items:

- Firewall CPU and memory
- Switch health and interface utilization
- Link aggregation (bond) status
- VLAN tagging errors
- Cluster health
- Storage replication status
- VM availability
- Internet latency

Alerting is configured for firewall or node outages, storage failures, link-aggregation failures, high resource utilization, and VLAN misconfiguration.

13. Security Controls

- Firewall administration restricted to the Management segment
- Unused switch ports disabled
- SSH only for administrative access; legacy unencrypted protocols disabled
- Strong authentication enforced for out-of-band management
- Firmware and software kept current
- Firewall logs reviewed regularly
- Storage-to-WAN deny rules reviewed monthly
- Out-of-band management access restricted to the Management segment
- Centralized logging / SIEM integration

14. Backup Procedure

Before any change, configuration backups are taken for the firewall, switches, hypervisor network configuration, and cluster configuration, and stored in a secure, access-controlled location.

15. Rollback Procedure

If a change fails validation: restore the firewall and switch configuration backups, restore the hypervisor network configuration, restart networking services, verify cluster health, and notify the SOC team.

16. Troubleshooting (Common Issues)

Issue	Resolution Approach
Hypervisor GUI unreachable	Verify Management VLAN and bond status
VM has no Internet access	Check VM VLAN and NAT policy
Storage failure	Verify Storage VLAN connectivity
Out-of-band management offline	Confirm Management-segment access
Storage traffic reaching WAN	Verify firewall rule ordering

Issue	Resolution Approach
Bond / link aggregation failure	Replace failed link and rebuild LACP

17. Change Management

All production changes follow a formal process:

- Raise a Change Request with implementation and rollback plans
- Obtain required approvals
- Schedule a maintenance window
- Notify the SOC team
- Execute the change
- Perform verification testing
- Update documentation

18. Maintenance Schedule

Frequency	Activity
Daily	Check firewall, switch, and cluster health
Weekly	Review logs and storage status
Monthly	Audit VLANs and firewall rules
Quarterly	Test failover and restore from backup
Annually	Review and update this SOP

19. Conclusion

This document outlines a standardized, portfolio-safe approach to designing, implementing, and operating a segmented VPS production network. By enforcing VLAN-based isolation, controlled access, redundancy, and structured change management, the architecture supports a secure, reliable, and highly available production environment. The complete operational SOP — including real addressing, port mappings, and vendor-specific detail — is maintained privately and shared only with authorized personnel.

Author

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